

1. Blume-Capel Model: Consider this extension of the Ising model on a d-dimensional lattice with z nearest neighbours.

$$H = -J \sum_{\langle i,j \rangle} S_i S_j + \Delta \sum_{i=1}^N S_i^2 - h \sum_{i=1}^N S_i$$

with $S_i = \{-1, 0, +1\}$. Here $\sum_{\langle i,j \rangle}$ denote a sum over nearest neighbours only, as usual.

- (a) Solve for the partition function in the mean field approximation.

Hint: Do not approximate the $(\Delta \sum_{i=1}^N S_i^2)$ term as this can be evaluated exactly. The free energy in mean field theory is

$$A = \frac{NJzm^2}{2} - k_B T N \ln \{1 + e^{-\beta \Delta} [2 \cosh(\beta [Jzm + h])]\},$$

where $m = \langle S_i \rangle$

- (b) Now expand the logarithm and write the free energy in Landau form with $h = 0$. What values of (T, κ) , where $\kappa = e^{-\beta \Delta}$ predict (a) normal critical behaviour, (b) tricritical behaviour, (c) a first-order transition? Find the value of the transition temperature in each case.
2. Determine the critical exponent λ for the following functions $\epsilon \rightarrow 0$. Here $\epsilon = (T - T_c)/T_c$.

$$f(\epsilon) = A\epsilon^{1/2} + B\epsilon^{1/4} + C\epsilon$$

$$f(\epsilon) = A\epsilon^{-2/3}(\epsilon + B)^{2/3}$$

$$f(\epsilon) = A(\epsilon^2 + B^2)^{1/2}(\ln |\epsilon|)^2$$

[Hint: If $f(\epsilon) \sim \epsilon^\lambda$, then $\lambda = \lim_{\epsilon \rightarrow 0} \frac{\ln f(\epsilon)}{\ln \epsilon}$.]

3. Problem 5.3, pg 165 Goldenfeld. [\[Click here to see the problem\]](#)

Hint for parts (c) and (d): Note that the system has three phases, high temperature phase ($M = 0$) and two low temperature phases ($M \neq 0$). Assume that the $M_{q_n} \neq 0$, $q_n \neq 0$ occurs only for $t < 0$ and treat the problem of minimization of M and q_n separately. If you are getting stuck, you can look up the paper referred to in the footnote but please follow Goldenfeld's notation and conventions.

4. First-order phase transitions: We discussed these in class but did not work out the algebra.

- (a) Consider the Landau free energy with a cubic invariant

$$\mathcal{L} = \frac{1}{2}\bar{a}_2(T - T_c)m^2 + \frac{1}{3}a_3m^3 + \frac{1}{4}a_4m^4$$

- i. Draw the free energy at different temperatures and convince yourself that there is indeed an abrupt transition
 - ii. Determine the transition temperature
 - iii. Determine the temperature window where the free energy has two coexisting minima
- (b) Now consider the Landau free energy with a sixth-order term

$$\mathcal{L} = \frac{1}{2}\bar{a}_2(T - T_c)m^2 + \frac{1}{4}a_4m^4 + \frac{1}{6}a_6m^6, \quad a_4 < 0$$

- i. Plot the free energy as a function of temperature and show that one has a symmetry-breaking abrupt (first-order) transition.
 - ii. Determine the transition temperature
 - iii. Determine the temperature window where the free energy has two coexisting minima
 - iv. Tricritical point: Show that we recover a continuous transition when $a_4 = 0$ and determine the critical exponents β , γ , and δ when $a_4 = 0$
- (c) Explain how and why you can get hysteresis? Can second-order transitions show hysteresis?
 - (d) For both the cases, try to understand the location of the minima and maxima of the Landau free energy for different temperatures, as discussed in class [see Arovas' notes].

Problem 4 need not be submitted.